

Chris Hickey

AI/BACKEND PYTHON ENGINEER

Seoul, South Korea

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SUMMARY

- Backend and AI engineer with ten years in production software, spanning telecoms test automation, unified communications, applied computer vision and, most recently, LLM conversation and retrieval systems serving live customers.
- Owns services end to end: architecture, asynchronous Python, data modelling, CI/CD, Kubernetes and observability.
- Primary reviewer for the retrieval and user-memory services at Brand Engagement Network.

PROFESSIONAL EXPERIENCE

Brand Engagement Network (BEN)

Seoul, Korea

AI ENGINEER

Jul. 2024 - Present

- Architected the shared asynchronous Python library behind all of the company's LLM dialog managers (generation, embeddings, emotion classification, conversation state). Decoupled speech synthesis from dialog logic across six customer assistants, letting multi-sentence responses be synthesised in parallel and cutting response latency.
- Designed and shipped a streaming chat API over WebSocket and SSE with typed event frames, automatic language switching, per-voice TTS routing and multilingual fallbacks. Brought the service to a typed, linted codebase with an 80% coverage gate in CI.
- Built the retrieval (RAG) service on PostgreSQL and pgvector: per-tenant databases, pluggable dense and sparse embeddings, background re-embedding and parallel query decomposition. Co-designed metadata-aware reranking and table and image retrieval from scientific papers.
- Led a coordinated breaking change replacing direct database access in five microservices with one typed data-service client, making schema ownership explicit and unblocking the multi-tenant migration.
- Added end-to-end trace IDs, Prometheus metrics and structured logging (Loki, Grafana) across the AI stack and owned its Kubernetes releases. Co-designed a user memory system that builds long-lived profiles from conversations via per-turn LLM extraction.

Surromind

Seoul, Korea

TEAM LEAD (PROMOTED FROM SENIOR AI RESEARCH ENGINEER, AUG. 2022)

Oct. 2021 - Jun. 2024

- Led government-funded computer vision projects: keypoint detection for classifying tennis strokes, and vision transformers for extracting descriptive image attributes such as shape, colour, depth and density.
- Built content-based music recommendation with CNNs and contrastive learning, plus CNN classifiers for instruments, mood and genre, extending the company's audio product line.
- Backend and machine learning engineer on AI Studio, a no-code AI platform. Wrapped open-source computer vision models (Mask R-CNN, YOLO) in TensorFlow and PyTorch behind Django, Redis and Celery services.

Blueface, A Comcast Business Company

Dublin, Ireland

SENIOR PYTHON DEVELOPER

Sept. 2020 - Oct. 2021

- Backend developer on Comcast's Unified Communications platform. Designed RESTful APIs with Flask and Marshmallow, and asynchronous task processing with Celery, RabbitMQ and Redis.

Common Computer

Seoul, Korea

BLOCKCHAIN DEVELOPER (PART-TIME)

Jan. 2019 - Sept. 2019

- Built a sharded, distributed database in Node.js on a Proof-of-Stake protocol, designed to be highly partitioned, always available and eventually consistent while maximising transactions per second.

Biointelligence Lab, Seoul National University

Seoul, Korea

GRADUATE RESEARCHER

Sept. 2018 - Sept. 2020

- Published research on hierarchical colour learning in CNNs, spoken command classification and motif discovery (see Publications), and edited lab members' publications on the Visual Turing Test.

Ericsson

Athlone, Ireland

TEAM LEAD

Sept. 2016 - Sept. 2017

- Led a three-person team building a version-controlled, automated continuous integration framework for Ericsson's Big Data Processing API, used across the company internationally.
- Automated deployment to a distributed server cluster for integration, functional and performance testing, and replaced manual environment setup with scripted DevOps tooling.

Ericsson

SOFTWARE ENGINEER

Athlone, Ireland

Sept. 2014 - Sept. 2016

- Built a mock network for load-testing Ericsson Network Manager, the company's flagship product, and a Selenium framework for testing its web interface.
- Led a refactor of the team's codebase on SOLID principles, improving performance while reducing the code to 30% of its original size.

EDUCATION

Seoul National University

Seoul, Korea

M. SC. ENGINEERING (COGNITIVE SCIENCE AND ARTIFICIAL INTELLIGENCE) · GPA 4.22/4.3

Sept. 2018 - Sept. 2020

University College Dublin

Dublin, Ireland

PROFESSIONAL DIPLOMA IN DATA ANALYTICS · GPA 4.0/4.2

Sept. 2015 - Sept. 2016

Athlone Institute of Technology

Athlone, Ireland

M. SC. APPLIED SOFTWARE ENGINEERING · FIRST CLASS HONOURS, TOP 10% OF CLASS

Sept. 2013 - Dec. 2014

University College Dublin

Dublin, Ireland

B. A. PSYCHOLOGY · GPA 3.84/4.2, 2ND IN A COHORT OF 70

Sept. 2009 - Sept. 2013

SKILLS

Languages

Python (main skill), TypeScript and Node.js · English (native), Korean (intermediate)

Backend

FastAPI, asyncio, Django, Flask, Celery, Redis, RabbitMQ, WebSockets and SSE, REST API design

AI & LLM

RAG pipelines, embeddings and reranking, pgvector, LLM evaluation (RAGAS), PyTorch, computer vision

Data & Infra

PostgreSQL, MongoDB, Docker, Kubernetes, Bitbucket Pipelines, semantic-release, Prometheus, Loki, Grafana

Practices

Test automation, code review, CI/CD, observability, technical writing

RESEARCH & PUBLICATIONS

- **Chris Hickey** and Byoung-Tak Zhang, "Hierarchical Color Learning in Convolutional Neural Networks", Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops 2020 (CVPRW 2020), Jun. 2020. pp.3484-3485.
- **Chris Hickey** and Byoung-Tak Zhang, "Classifying verbal commands using Convolutional Neural Networks and Mel-spectrographic sound representations", Proceedings of Korea Computer Congress 2020 (KCC 2020), Jul. 2020.
- **Chris Hickey** and Byoung-Tak Zhang, "Split Gibbs Sampling Algorithm for Consistent Motif Discovery", Proceedings of Korea Computer Congress 2020 (KCC 2020), Jul. 2020.
- Sungjae Cho, Jaeseo Lim, **Chris Hickey**, Jung Ae Park, and Byoung-Tak Zhang, "Simulating problem difficulty in arithmetic cognition through dynamic connectionist models", Proceedings of the 17th International Conference on Cognitive Modelling (ICCM 2019), Jul. 2019. pp.29-34.
- Sungjae Cho, Jaeseo Lim, **Chris Hickey**, and Byoung-Tak Zhang, "Problem difficulty in arithmetic cognition: Humans and connectionist models", Proceedings of the 41st Annual Meeting of the Cognitive Science Society (CogSci 2019), Jul. 2019. pp. 1506-1512.

SCHOLARSHIPS, VOLUNTEERING & ADDITIONAL INFORMATION

- **Korean Government Scholarship (KGSP), 2017 - 2020.** Sole Irish recipient; full tuition for the M. Sc. at Seoul National University.
- **Ericsson Scholarship, 2013 - 2014.** Full tuition and stipend for the M. Sc. at Athlone IT, tied to joining Ericsson on graduation.
- **English teaching, 2011 - 2018.** Two years teaching in Zambia, Hong Kong, Vietnam and Ireland; CELTA certified (2011).
- **Samaritans phone counsellor, 2015 - 2017.** Night-shift volunteer on a listening hotline for people in Ireland in distress.